

Introduction



In today's digital world, distinguishing real from AI-generated face images has become a serious challenge. As artificial intelligence advances, so do the tools for creating hyper-realistic fake identities, posing significant risks to personal privacy, cybersecurity, and public trust.



Figure 1. Identity theft

Identity theft affects key sectors. Despite the rise of biometric authentication systems, attackers have found ways to bypass them using convincingly forged facial images. The result? A growing need for robust, reliable methods to verify whether a face is authentic or artificially generated.

This work addresses that need by introducing a novel approach based on Image Quality Assessment (IQA) techniques. Rather than comparing faces to existing databases, our method analyzes the intrinsic visual quality of the image itself to detect signs of AI synthesis—offering a fast, standalone verification tool.

Problem & Motivation

- **The Rise of AI-Generated Faces :** Recent advances in generative models (GANs, diffusion models) have made it easy to create synthetic facial images that are nearly indistinguishable from real ones. These fake identities are now used in fraud, misinformation, and identity theft.
- **Limitations of Traditional Biometric Systems :** Despite their growing adoption, biometric authentication systems remain vulnerable to high-quality forgeries. Criminals can bypass facial verification using AI-generated images, undermining the trust placed in these technologies.
- **Need for Quality-Based Verification :** Instead of relying on identity comparison with a known database, analyzing intrinsic visual quality offers a promising solution. No-Reference Image Quality Assessment (NR-IQA) techniques can reveal subtle artifacts left by generative models, offering a fast, standalone method to assess authenticity.



Figure 2. Example of a real vs. AI-generated face

Related Work

State-of-the-Art in detecting real and AI-Generated face image

Traditional Feature-based methods

- ✓ Process visible spectrum images
- ✓ Extract specific facial features (texture, edges, shape)
- ✓ Look for imperfections in lighting, eyes, teeth, facial contours

→ **Limitation:** *Less effective against advanced forgery techniques*

Deep Learning-based methods

- ✓ Convolutional Neural Networks (CNNs)
- ✓ Multi-stage deep learning approaches
- ✓ Pre-trained architectures (VGG16, ResNet50) with fine-tuning

→ **Limitation:** *Performance decreases as forgery techniques improve*

Explainable AI Techniques

- ✓ Prioritizes explainability alongside accuracy
- ✓ Uses Gradient class activation Mapping
- ✓ Focuses on subtle imperfections in backgrounds

→ **Advantage:** *Provides insight into decision-making process*

Multispectral imaging methods

- ✓ Uses multiple spectral bands beyond visible spectrum
- ✓ Captures complementary information from different wavelengths
- ✓ Merges information to improve detection accuracy

→ **Limitation:** *Increased complexity in capture devices*



Emerging forgery techniques reduce accuracy of current detection methods

Figure 3. Review of state-of-the-art methods

The Proposed Facial Image Authenticity Verification (FIAV) Method

FIAV (Face Image Assessment for Verification) assesses image quality features to differentiate real from AI-generated face images. The key steps of FIAV method are :

1. **Input: Facial Image** Accepts any facial image (real or AI-generated) for analysis.
2. **Feature Extraction :** Leverages 4 advanced NR-IQA metrics (MANIQA, NIQE, IL-NIQE, TOPIQ) + 4 basic metrics (blur, contrast, brightness, sharpness).
3. **Feature Selection :** Recursive Feature Elimination (RFE) optimizes feature relevance.
4. **Classification Model :** Random Forest/SVC trained on 2,097 images (balanced real/AI).
5. **Output: Real/Fake Decision :** Binary output: Real (label '1') / AI-generated (label '0').

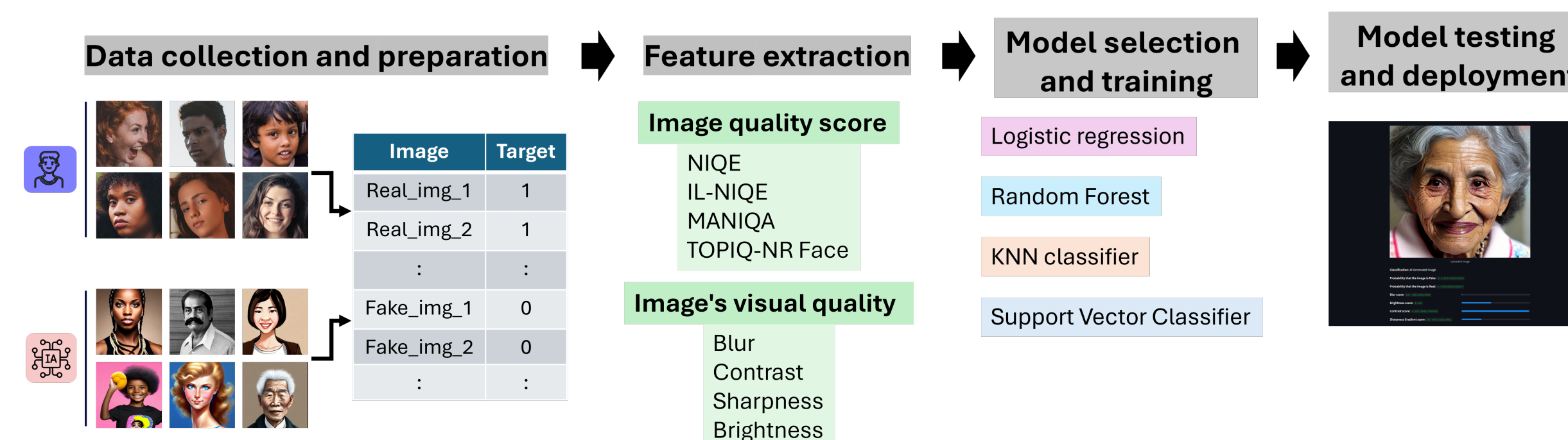


Figure 4. FIAV method

Deployment via Streamlit

A fully functional Python web interface is proposed. It allows users to:

- Upload face images
- Trigger classification
- Instantly view results
- Enhances accessibility and supports real-world usability

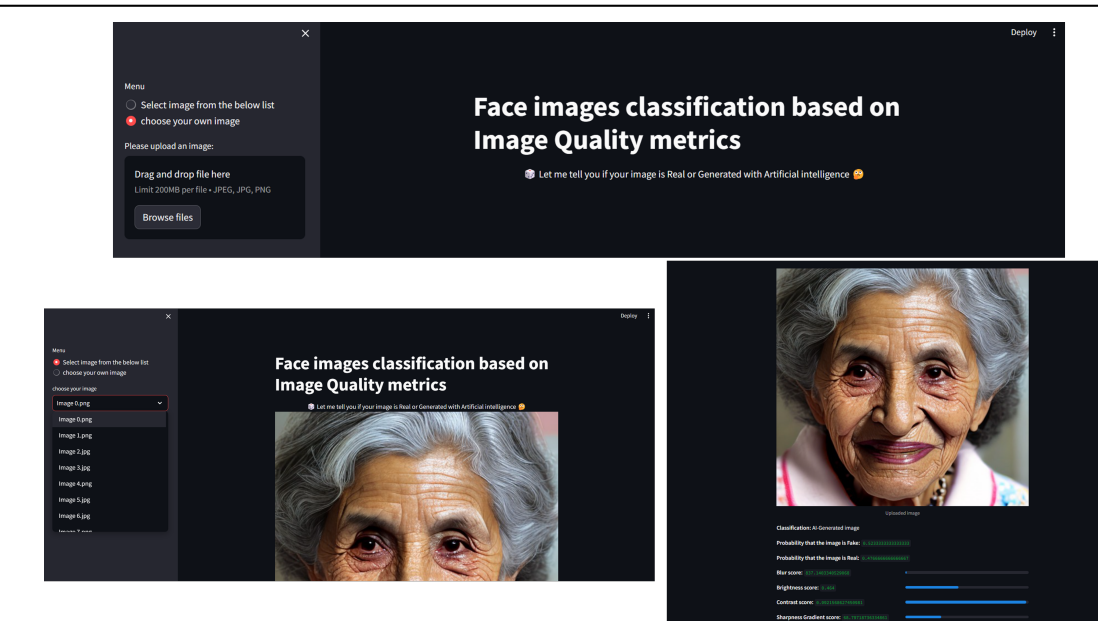


Figure 5. Web interface

Experimental Setup and Results

Datasets and Protocol

- **Datasets:** 1,016 real facial images and 1,081 AI-generated images.
- **Split:** 80% training / 20% testing (stratified).
- **Evaluation metrics:** Accuracy, F1-score, Confusion Matrix.
- **Features:** Selected via RFE: *MANIQA, Contrast, NIQE, Brightness, TOPIQ-NR_FACE, Blur, Sharpness.*

Classifier Settings

- **Logistic Regression:** max_iter=400
- **Random Forest:** max_depth=30, min_samples_leaf=1, n_estimators=300
- **KNN:** n_neighbors=6
- **SVM:** RBF kernel

Performance Summary

	Logistic Regression	Random Forest
Actual 0	163	46
Actual 1	34	178
Predicted 0	184	25
Predicted 1	21	191

(a) (b)

	KNN	SVC
Actual 0	162	47
Actual 1	79	133
Actual 0	169	40
Actual 1	76	136

(c) (d)

Table 1. Accuracy and F1-score of classifiers

Model	Accuracy	F1-score
Logistic Regression	82.42%	82.71%
+ Feature Selection	81.95%	81.73%
Random Forest	89.78%	89.63%
+ Feature Selection	87.17%	86.82%
KNN	70.07%	67.85%
SVM	71.02%	70.10%

Table 2. Accuracy using individual and combined features

Model	Best Feature	Accuracy	FIAV Accuracy
Logistic Regression	NR-IQA (M9)	75.53%	82.42%
Random Forest	IVQ (M10)	82.18%	89.78%
SVM	Contrast (M7)	74.82%	71.02%

Figure 6. Confusion matrices for (a) Logistic Regression, (b) Random Forest, (c) KNN, (d) SVM

- FIAV combines NR-IQA + IVQ scores.
- **Best results with Random Forest (FIAV: 89.78%).**
- Features like *MANIQA* and *IVQ* show strong discriminative power.

Conclusion

- **Random Forest + FIAV = Best overall performance.**
- Feature fusion significantly improves classification.
- Future work: investigate deeper models.